Retail Sales Performance and Profitability Analysis using SQL and Power BI

1. Business Objective

This project evaluates retail sales performance and how effectively the business converts revenue into profit. The analysis was designed as a business case study rather than a basic SQL exercise, with a focus on growth, product performance, customer value, regional variation, discount impact, and profitability risk.

The main questions addressed were whether the business is growing over time, which categories and products create the most value, whether high sales always translate into strong profit, which customers and markets are most valuable, and how discounting affects margin performance. A further aim was to identify whether profit is concentrated in a small number of products or customers and where the business is losing money.

2. Dataset Overview and Validation

The dataset is recorded at **transaction line level**, meaning each row represents one product line within an order rather than one complete order. This is important because one order_id can appear across multiple rows, so order based metrics had to be calculated using distinct order counts.

The dataset contains **9,994 rows**, **5,009 distinct orders**, **793 customers**, **3 categories**, and **17 sub categories**, covering the period from **2014 to 2017**. Validation checks confirmed that row_id is unique, key analytical fields contain no missing values, and no records were found where ship_date occurs before order_date. Numeric ranges were commercially plausible, with discounts ranging from **0.00 to 0.80** and profit ranging from strong gains to substantial losses.

Metric	Value
Total Sales	2,297,201.07
Total Profit	286,397.02
Total Orders	5,009
Total Customers	793
Total Quantity	37,873
Average Order Value	458.61
Profit Margin	12.47%

Table 1: Core KPI Summary

Overall, the business is profitable, but later analysis shows that profitability is not evenly distributed across categories, products, customers, or geographies.

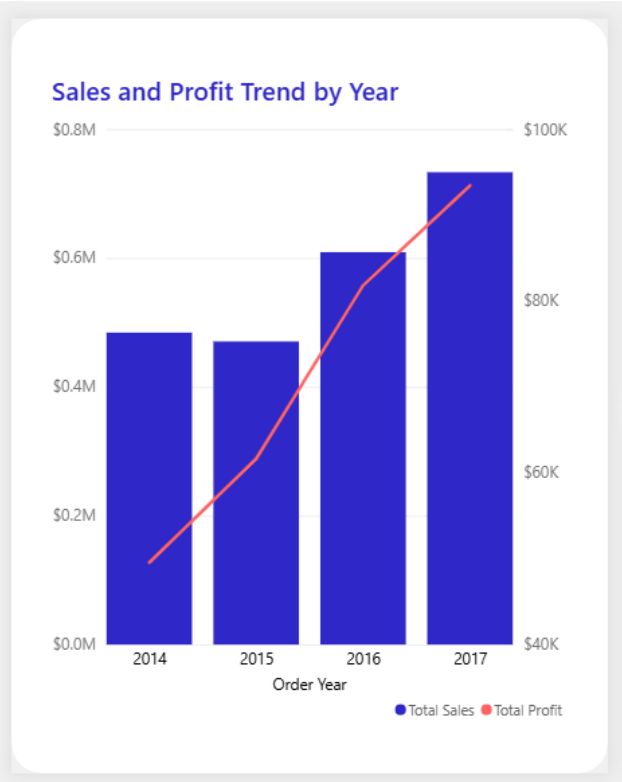
3. Growth and Overall Performance

The business grew overall during the period, although growth was not perfectly linear. Sales increased from **484.2 thousand** in 2014 to **733.2 thousand** in 2017, while profit rose from **49.5 thousand** to **93.4 thousand**. Order volume also increased steadily, showing that growth was supported by rising transaction activity.

However, revenue and profit did not always move together. In 2015, sales declined slightly while profit rose strongly, suggesting improved margin quality. In 2016, both sales and profit grew strongly, making it the strongest all-round year. In 2017, sales continued to grow, but profit growth slowed relative to revenue, indicating some weakening in growth quality.

Year	Sales	Profit	Orders	Profit Margin
2014	484,247.56	49,543.97	969	10.23%
2015	470,532.46	61,618.60	1,038	13.10%
2016	609,205.86	81,795.17	1,315	13.43%
2017	733,215.19	93,439.27	1,687	12.74%

Table 2: Yearly Performance



Insert Visual 1: Sales and Profit Trend by Year

4. Product and Customer Performance

Product performance is uneven across the business. At category level, **Technology** and **Office Supplies** are strong value drivers, both delivering margins above **17 percent**. **Furniture** is the weakest category, producing substantial revenue but only a **2.49 percent** margin.

Within Furniture, **Tables** and **Bookcases** emerged as major structural weak points. Tables generated more than **206.9 thousand** in sales but produced a loss of **17.7 thousand**. By contrast, sub categories such as **Copiers**, **Paper**, **Labels**, and **Envelopes** were strong value creators.

Customer analysis showed a similar pattern. High customer spend did not always translate into high value. The highest revenue customer in the dataset was loss making overall, while several other customers generated lower sales but much stronger profit margins. Profit also proved more concentrated than sales: the top 10 products by profit generated **23.21 percent** of total profit from only **8.36 percent** of sales, and the top 10 customers by profit contributed **17.52 percent** of total profit from only **5.33 percent** of sales.

At segment level, **Consumer** is the main volume engine of the business, but it has the weakest margin. **Home Office** is smaller in scale but the most efficient segment, while **Corporate** performs better than Consumer on both margin and average order value.

Area	Sales	Profit	Profit Margin
Technology	836,154.10	145,454.95	17.40%
Furniture	741,999.98	18,451.27	2.49%
Office Supplies	719,046.99	122,490.80	17.04%
Consumer	1,161,401.34	134,119.21	11.55%
Corporate	706,146.44	91,979.13	13.03%
Home Office	429,653.29	60,298.68	14.03%

Table 3: Category and Segment Summary

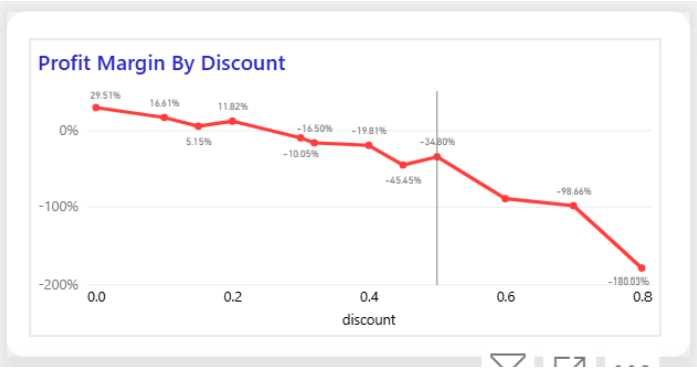
5. Regional and Discount Risk

Geographic analysis shows that performance varies significantly across markets. **West** is the strongest region overall, followed by **East**, while **Central** is the weakest region by margin. At state level, **California** and **New York** are the main value centres, combining large sales with strong profitability. In contrast, **Texas** is the largest geographic profit drain, despite being one of the highest revenue states. Other major loss making states include **Ohio**, **Pennsylvania**, and **Illinois**.

Discounting emerged as one of the clearest drivers of weak profitability. Transactions with no discount generated the strongest margin at **29.51 percent**, while profitability turned negative from around **30 percent discount** onward. The effect was not uniform across categories. **Furniture** was the most discount sensitive category, with **Tables** and **Bookcases** becoming unprofitable at relatively modest discount levels.

Area	Key Result
West	Strongest region overall
Central	Weakest regional margin
California	Major value centre
New York	Major value centre
Texas	Largest state level profit drain
Ohio	Major loss making market
Pennsylvania	High sales but negative profit
Illinois	Major underperforming state
Discounts above 30%	Profitability turns negative
Furniture	Most discount sensitive category
Tables and Bookcases	Structural weak points

Table 4: Regional and Risk Summary



Visual 2: Profit Margin by Discount Level

6. Key Findings and Recommendations

The analysis highlights six core findings. The business is profitable overall, but profitability is uneven across the portfolio. Growth is positive over time, although sales and profit do not always move together, showing that revenue growth alone is not a reliable measure of performance quality. Technology and Office Supplies are the strongest value drivers, while Furniture is structurally weak, particularly in Tables and Bookcases. The analysis also shows that high sales do not always translate into high value, as some top revenue products and customers are loss making. In addition, profit is more concentrated than sales across both products and customers, creating some dependence on a relatively small group of strong performers. Finally, discounting is a major driver of margin erosion, especially in Furniture and in weak states such as Texas, Ohio, Pennsylvania, and Illinois. Overall, SQL and Power BI helped identify where value is created, where it is lost, and where the business faces commercial risk.